

# Diagonality

## The Best of Both Worlds

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Testing the tactical theory of diagonality — and turning it into a real-time, orientation-aware map — from 3D skeleton tracking.

Jaime Oriol



# DIAGONALITY: THE ORIENTATION-AWARE PIPELINE

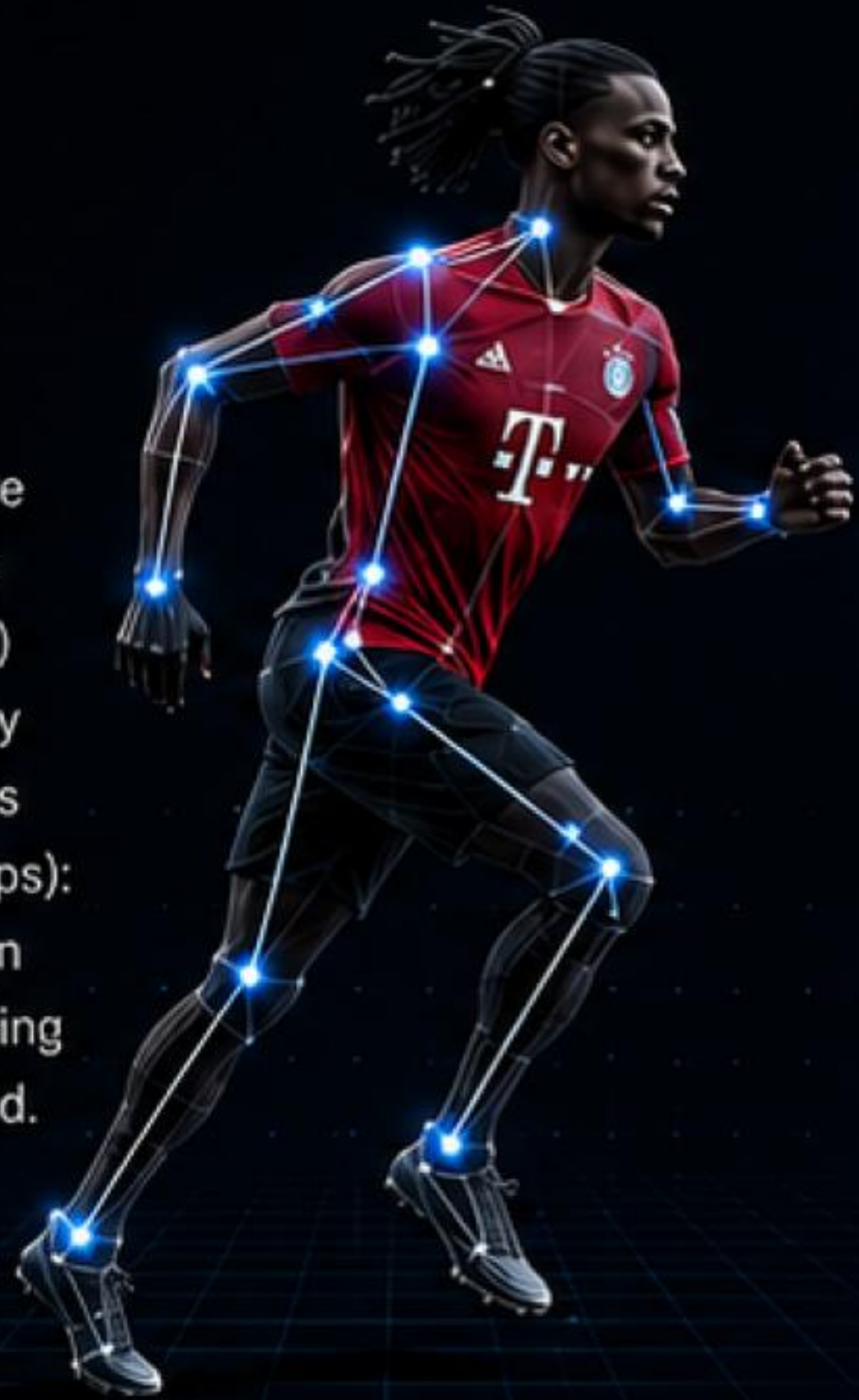
FROM 3D SKELETON TRACKING TO A REAL-TIME MAP OF DIAGONAL OPPORTUNITY

• AWS WORLD SPORTS INNOVATION CUP 2026 • CHALLENGE 2 •

## INPUT 1 · 3D SKELETON

21 body  
keypoints  
at 50 Hz.

We read where  
each player is  
looking (head)  
and which way  
the body faces  
(shoulders, hips):  
the orientation  
ordinary tracking  
never captured.



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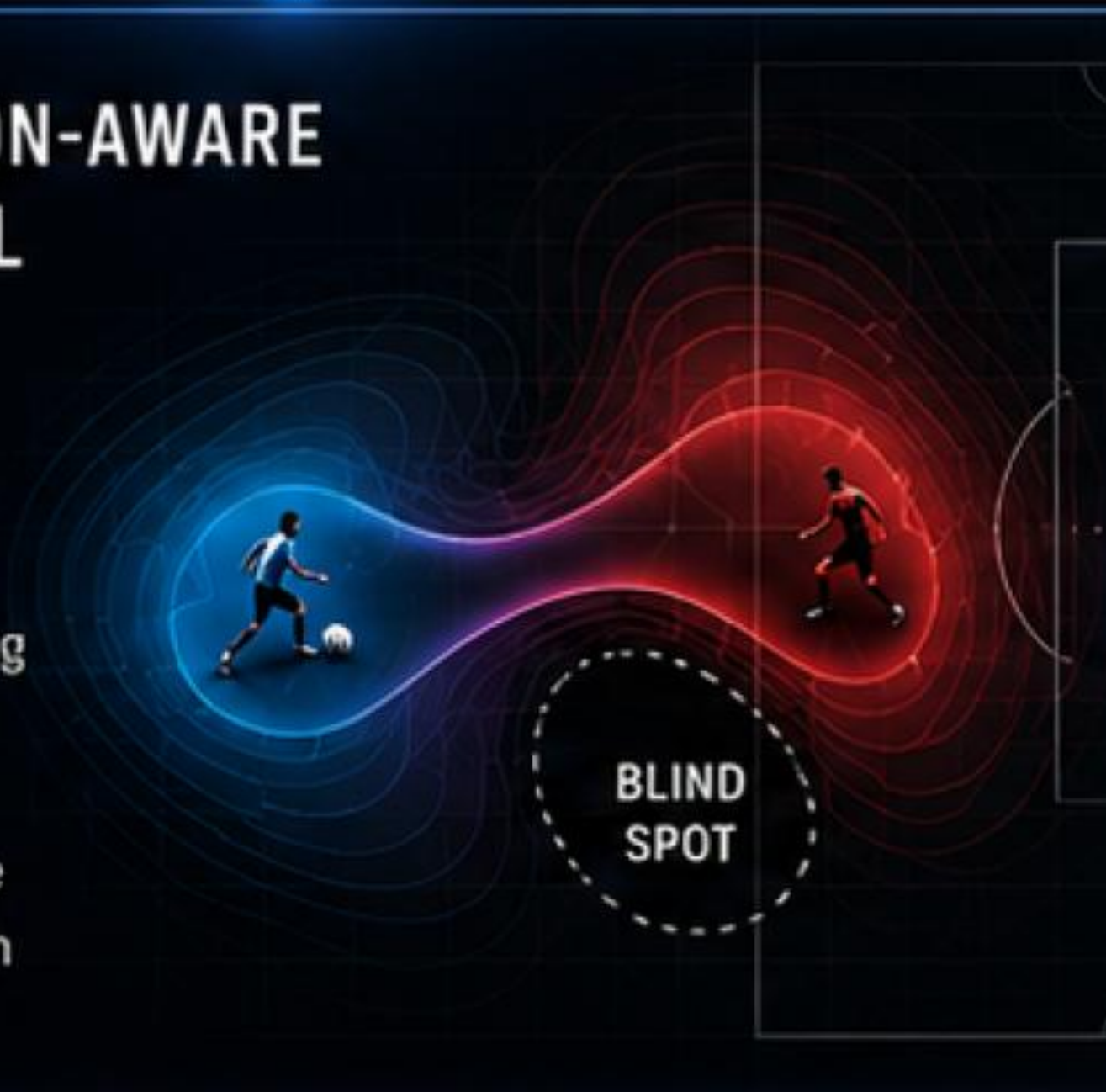
## 1 · VISION

What every defender  
can actually see.  
  
A probabilistic 120°  
field-of-view cone  
around the real head  
direction. It narrows  
as the player runs  
faster, and other  
players cast occlusion  
shadows.

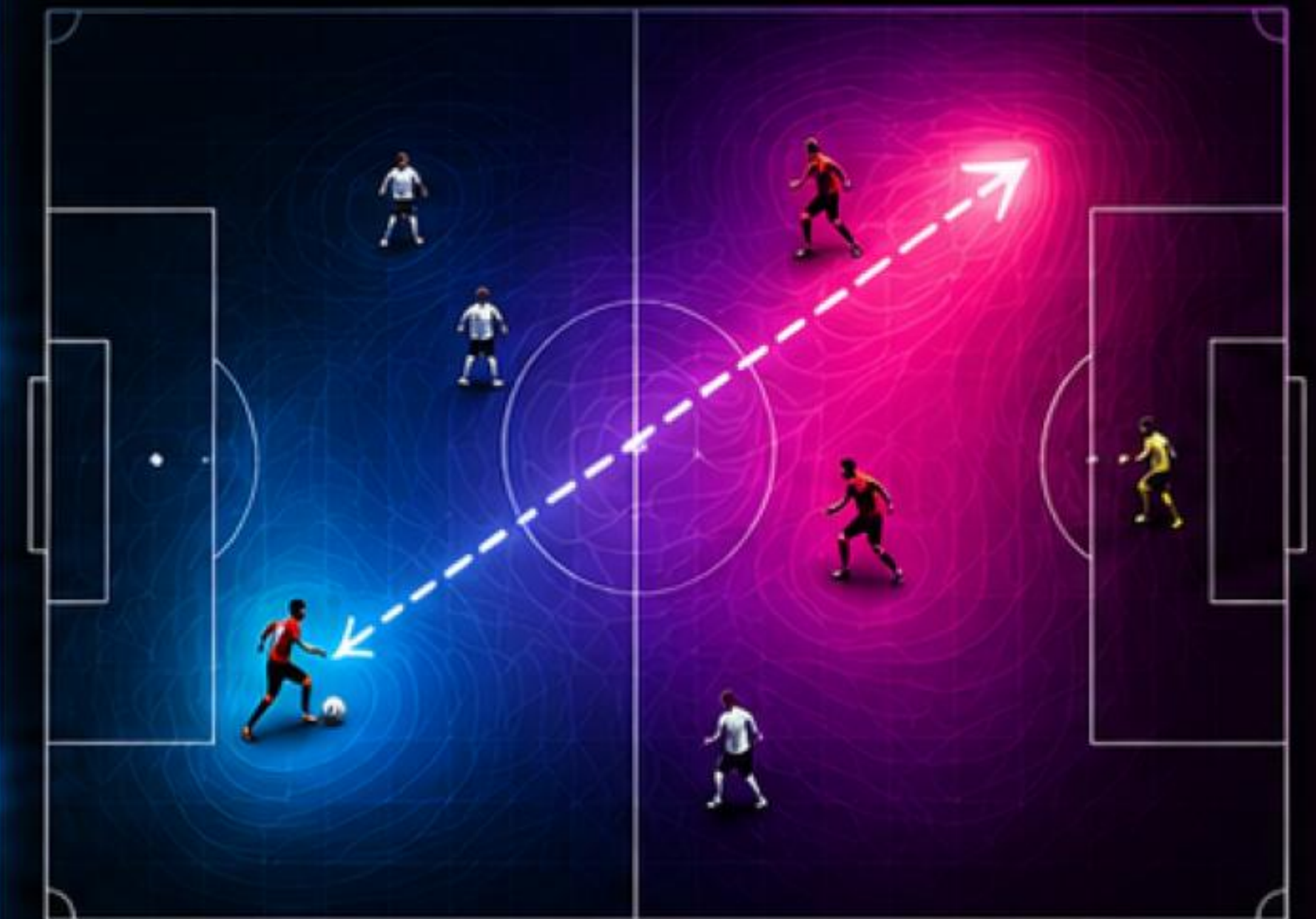


## 2 · ORIENTATION-AWARE PITCH CONTROL

Not plain pitch control.  
What every player can  
reach given which way  
they are turned: reaching  
for a threat behind the  
shoulder costs reaction  
plus turning time, so the  
control blob collapses in  
the blind spot.



## 3 · DOS: DIAGONAL OPPORTUNITY SURFACE



Vision and pitch control fused. The extra control an  
attacker buys by delivering the ball diagonally (into  
a defender's blind side) instead of straight. A blind  
defender reacts late, his reach collapses, and DOS  
lights up the gap.



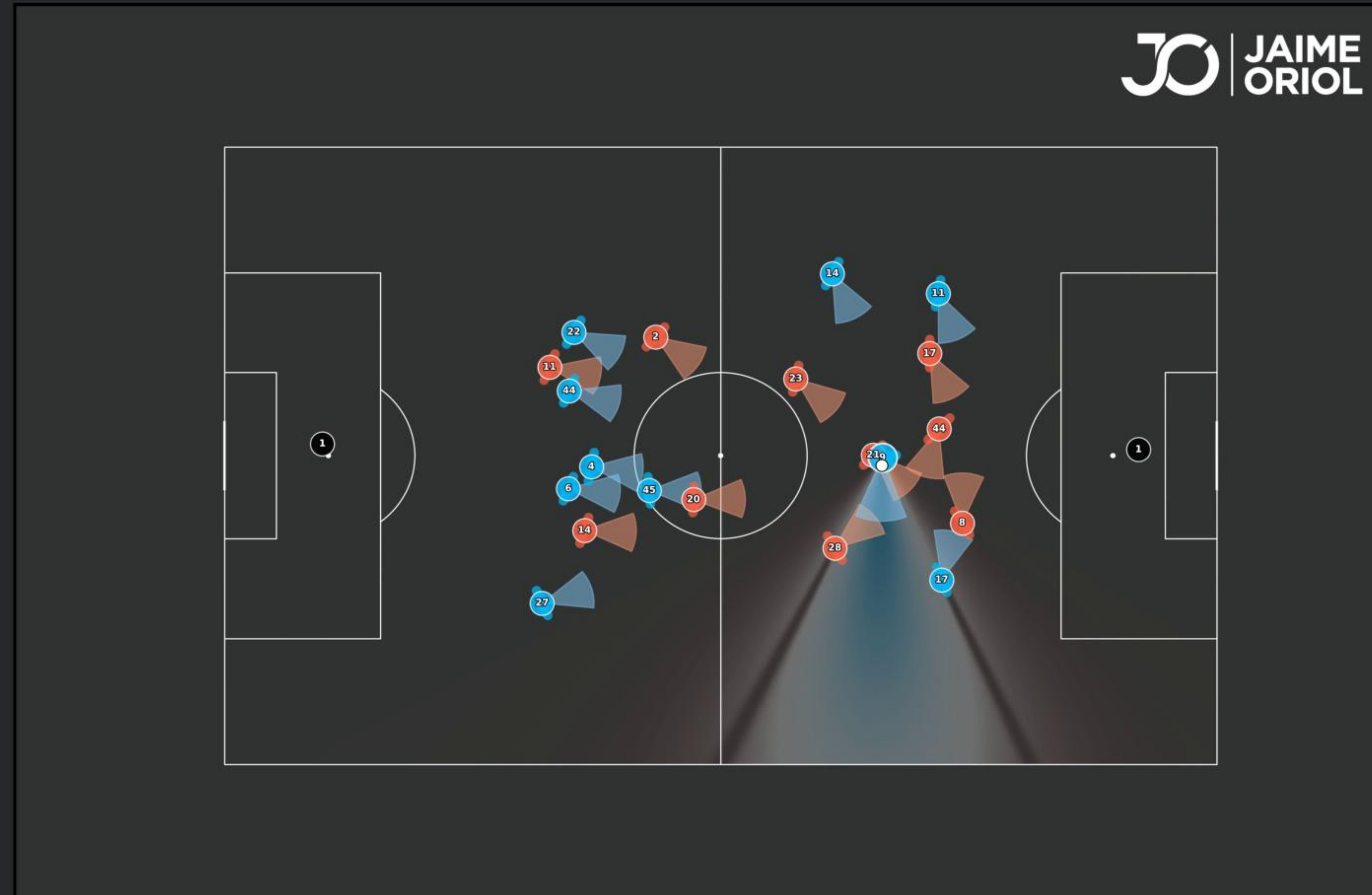
TRACAB 3D SKELETON

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CREATED BY JAIME ORIOL



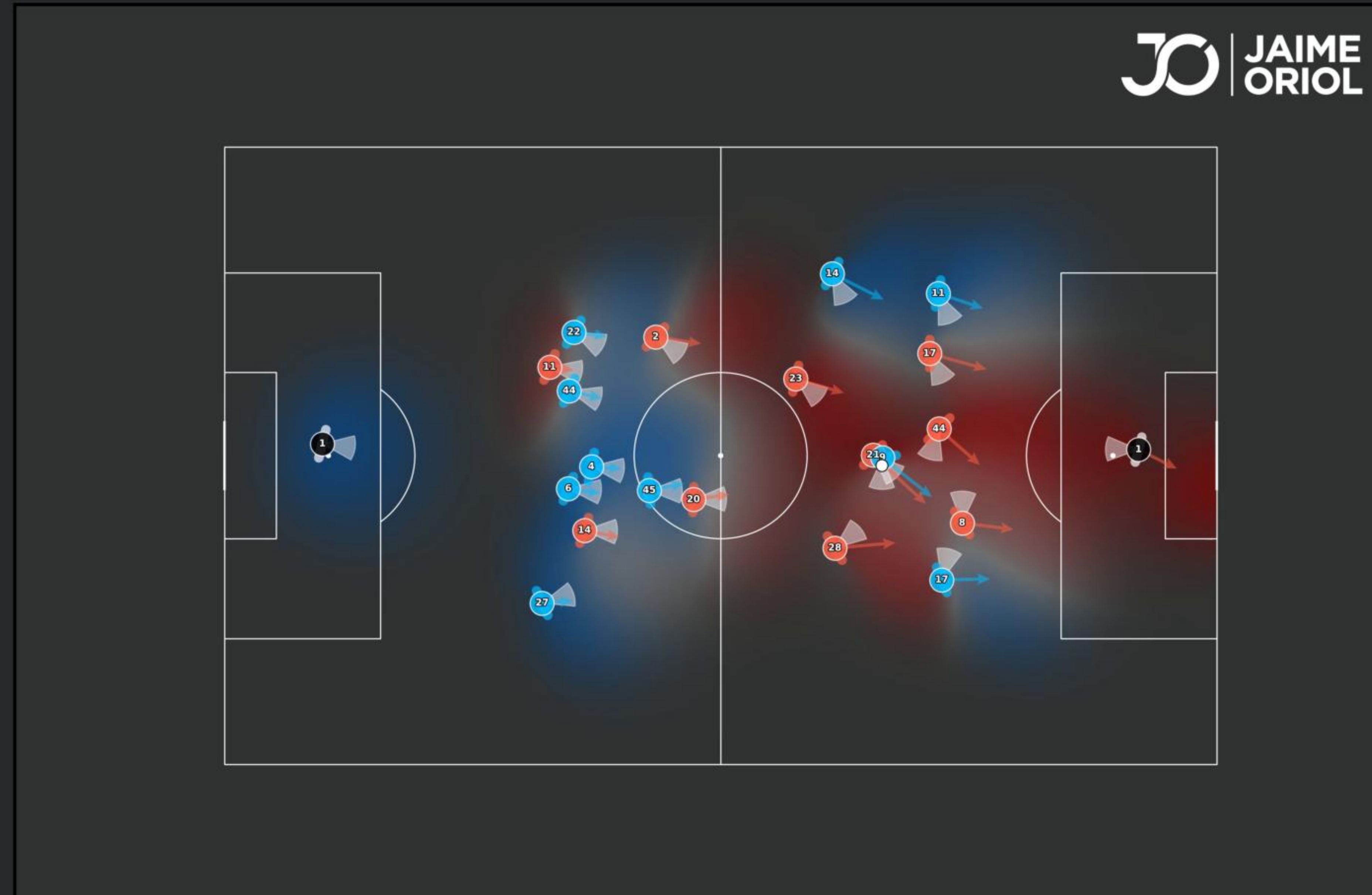
# What every player can see



Probabilistic field of view: a 120° cone on the measured head yaw, speed-dependent decay, torso occlusion from real shoulder widths.



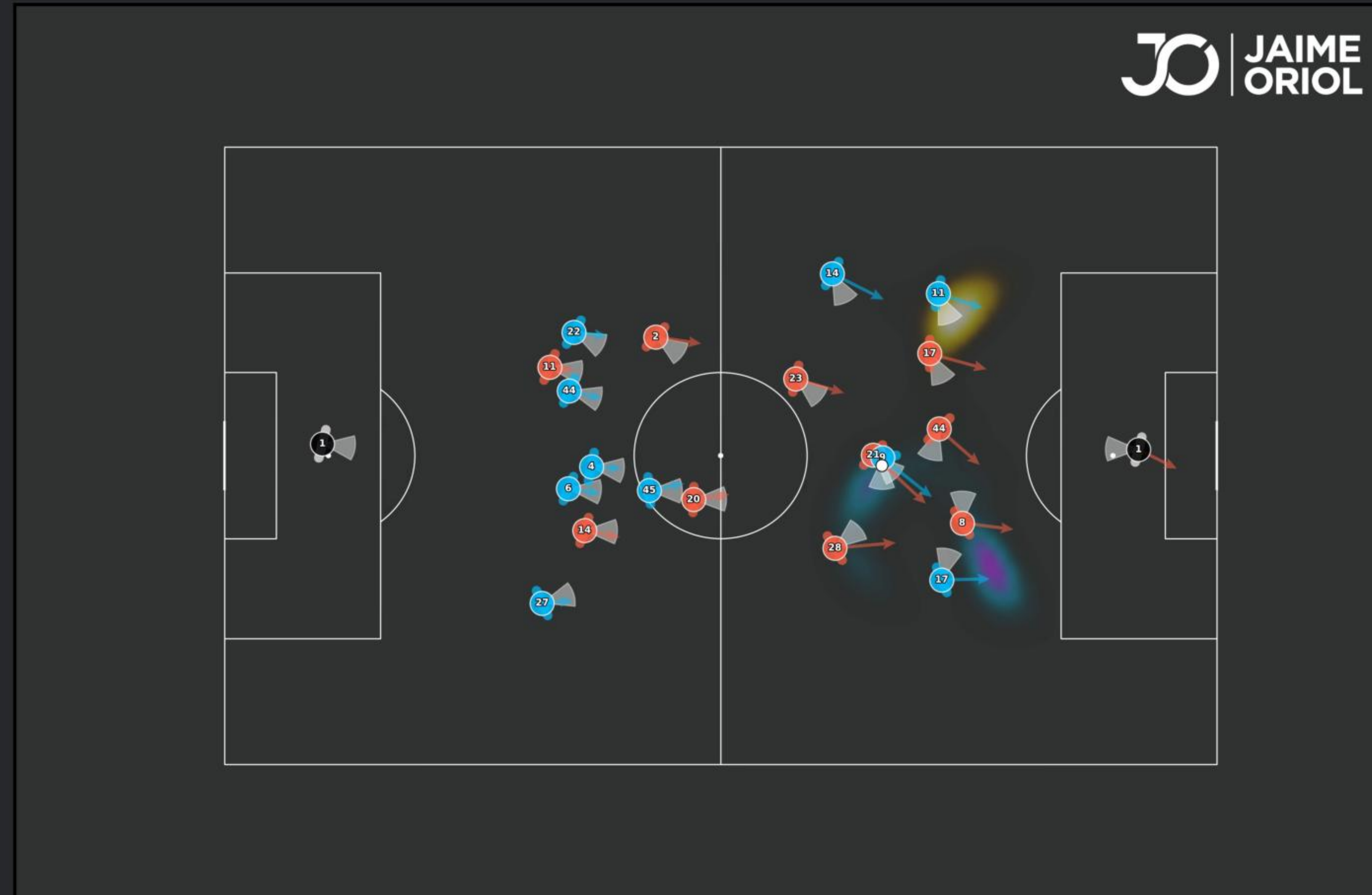
# What every player can reach



Each player an anisotropic reach field — the blob collapses in a defender's blind spot, a literal hole in his control of space.



# The Diagonal Opportunity Surface



Extra control bought going diagonal vs straight, gated to what the on-ball player perceives. Cyan: seen now. Amber: opportunity lost track of.